

ASSIGNMENT 13

ROLL NO:CSB24083

NAME:DIPANKAR BEZBARUAH

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
    char *data;
    size_t length;
    size_t capacity;
} StringBuffer;

StringBuffer* sb_init(size_t initial_capacity) {
    StringBuffer *sb = malloc(sizeof(StringBuffer));
    if (sb == NULL) {
        fprintf(stderr, "Struct allocation failed\n");
        return NULL;
    }

    sb->data = malloc(initial_capacity * sizeof(char));
    if (sb->data == NULL) {
        fprintf(stderr, "Buffer allocation failed\n");
        free(sb);
        return NULL;
    }

    sb->length = 0;
    sb->capacity = initial_capacity;
    sb->data[0] = '\0';
```

```

    return sb;
}

int sb_append(StringBuffer *sb, const char *str) {
    if (sb == NULL || str == NULL)
        return 0;

    size_t str_len = strlen(str);

    while (sb->length + str_len + 1 > sb->capacity) {
        size_t new_capacity = sb->capacity * 2;

        char *temp = realloc(sb->data, new_capacity *
sizeof(char));
        if (temp == NULL) {
            fprintf(stderr, "Realloc failed\n");
            return 0;
        }

        sb->data = temp;
        sb->capacity = new_capacity;

        printf("Resized to: %zu\n", sb->capacity);
    }

    memcpy(sb->data + sb->length, str, str_len + 1);

    sb->length += str_len;

    return 1;
}

void sb_free(StringBuffer *sb) {
    if (sb == NULL)

```

```
        return;

    free(sb->data);
    free(sb);
}

int main() {
    StringBuffer *sb = sb_init(4);

    sb_append(sb, "Hi");
    sb_append(sb, " ");
    sb_append(sb, "there!");
    sb_append(sb, " This will grow again.");

    printf("\nFinal: %s\n", sb->data);
    printf("Length: %zu\n", sb->length);
    printf("Capacity: %zu\n", sb->capacity);

    sb_free(sb);
    return 0;
}
```

Output:

Resized to: 8

Resized to: 16

Resized to: 32

Final: Hi there! This will grow again.

Length: 31

Capacity: 32